

NAME: _____ ID: _____

CSC 343 H5S

Day Class 01

Instructor: Michael Liut

DURATION: 2 hours

University of Toronto

Midterm Examination

March 5th, 2018

Read all instructions before starting the exam

This test paper includes 11 pages and a total of 14 questions; 10 true-or-false questions and 4 questions requiring a written answer. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions :

1. The answers to the written questions (Questions 11-[14/15]) must be answered in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked.
2. Read all questions before starting the exam – some are easier than others.
3. The use of the CASIO FX-991 is permitted, but highly discouraged. No questions herein require a calculator.
4. **This is a closed book exam.** No memory aids, notes, or textbooks of any kind are allowed during the test. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
5. The students are not allowed to be involved in any communication of any kind concerning the questions and answers of this exam with anybody except the invigilator(s) or the instructor. Any attempt of such communication will be considered a case of academic dishonesty.
6. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked.
7. Documents to be returned: this questionnaire, the cheat sheet and all scrap paper if used. All of these documents must bear the student's name and number. Only the face page of the questionnaire needs to bear the student's name and number.

question(s)	mark	out of
1 – 10		15
11		6
12		3
13		3
14/15		3
misc.		n/a
total		30

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Question 1 – 10 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example.

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word “True” or the word “False”. The negative marking scheme is not used. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. You must justify your answer.

I. General Knowledge (15 marks)

Question 1 [1.5 marks] A relation $R(A,B,C)$ may have at most one (minimal) key (not superkeys).

- A. True.
- B. False.

Question 2 [1.5 marks] A weak entity-set is an entity-set that cannot be uniquely identified by its attributes alone; therefore, it must use its primary key in conjunction with a foreign key to uniquely identify tuples.

- A. True.
- B. False.

Question 3 [1.5 marks] When speaking of referential integrity, the dependant table is denoted by the creation of a Foreign Key in a table. The table being referenced would, therefore, be the parent.

- A. True.
- B. False.

Question 4 [1.5 marks] Let relation R be a bag over the attributes A,B . Now let's say that A is our primary key for the relation R . Thus, R is necessarily a set.

- A. True.
- B. False.

Question 5 [1.5 marks] A weak entity-set **E** has a relationship with a supporting entity-set **R**. The relationship between **E** and **R** must be one-to-one.

- A. True.
- B. False.

Question 6 [1.5 marks] Arity is the number of arguments/operands that a function takes in. Cardinality is a measure of the number of elements of the set. More specifically, cardinality is the number of rows and degree/arity is the number of columns in a relation's instance.

- A. True.
- B. False.

Question 7 [1.5 marks] Set union is idempotent, i.e. $S \cup S = S$; while under bag law it is not, i.e. $S \cup S \neq S$. Thus, in general, it is safe to assume that intersection is also NOT idempotent; i.e. $S \cap S \neq S$.

- A. True.
- B. False.

Question 8 [1.5 marks] Given two relations, $R(A,B)$ and $S(B,C)$, $\sigma_{A=B}(R \times S) = (R \bowtie_{A=B} S)$.

- A. True.
- B. False.

Question 9 [1.5 marks] Given two relations, **Emps** and **WorksIn**, if the following SQL Query is executed, then its result is undoubtedly the **Output** relation.

Emps:

Emp	Sal	Mgr
Xin	100	Pat
Jen	200	Jen
Pat	100	Meg
Meg	150	Meg
Joe	110	Ami
Ami	200	Bob

WorksIn:

Dept	Emp
Toy	Xin
Car	Bob
Car	Meg
Art	Pat
Art	Meg

SQL Query

```
SELECT *
FROM Emps,
    (select Emps.emp, max(sal) as sal
     from Emps, WorksIn
     where Emps.emp = WorksIn.emp
     group by Emps.emp) as foo
WHERE Emps.mgr = foo.emp
```

Output:

Emp	Sal	Mgr	Emp	Sal
Xin	100	Pat	Pat	100
Pat	100	Meg	Meg	150
Meg	150	Meg	Meg	150

- A. True.
- B. False.

Question 10 [1.5 marks] Given the two relations in Question 9, **Emps** and **WorksIn**, if the following SQL Query is executed, then its result is undoubtedly the **Output** relation.

SQL Query

```
SELECT E1.Emp
FROM Emps E1 full outer join Emps E2 on E1.Emp = E2.Mgr
GROUP BY E1.Emp, E2.Emp
HAVING sum(E1.Sal) >= sum(E2.Sal)
ORDER BY E1.Emp
```

Output:

Emp
Ami
Jen
Meg
Meg
Pat

- A. True.
- B. False.

Questions 11 – 13 are questions that require a written answer (some in the form of an SQL query). The answers are to be provided in the space provided below each question.

II. SQL (12 marks)

Question 11 [6 marks] Consider the schema below about movies and actors. Keys are underlined.

- Actor(aid, name)
- Movie(mid, title, year, genre, length)
- Roles(aid, mid, character)

Write an SQL query which finds all actors who did not star in any movie (i.e. play a character) in the year 2000. Only return the actor's name and actor ID.

Question 12 [3 marks] Consider the following CREATE TABLE definition:

```
CREATE TABLE Midterm(  
    A INT NOT NULL,  
    B INT NOT NULL,  
    C INT NOT NULL,  
    PRIMARY KEY (A),  
    FOREIGN KEY (B) REFERENCES Midterm(A) ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY (C) REFERENCES Midterm(A) ON DELETE CASCADE ON UPDATE RESTRICT  
);
```

Now consider that the table `Midterm` has the following entries:

A	B	C
3	3	3
4	3	3
5	4	3

What is the result of the following SQL statement:

```
DELETE  
FROM Midterm  
WHERE A = 3
```

Question 13 [3 marks] Recall the **Cartesian Product** of two relations R and S , denoted $R \times S$, where R and S are sets: $R \times S = \{(r, s) : (r \in R) \text{ and } (s \in S)\}$. Is the Cartesian product operation commutative? That is, in general, is $(R \times S) = (S \times R)$? Prove/Disprove.

Questions 14 – 15 are questions that involve the translation of SQL queries into relational algebra. Each of these questions relate to the following schema:

- Suppliers(sid:integer, sname:string, address:string)
- Parts(pid:integer, pname:string, colour:string)
- Catalog(sid:integer, pid:integer, cost:real)

Please note, in the schema above, keys for each relation are denoted by an underlined. Additionally, the Catalog relation lists the prices charged for parts by Suppliers (i.e. cost).

III. Relational Algebra (3 marks)

Pick only one to answer: Question 14 or Question 15. Do not answer both.

Question 14 [3 marks] Given the following SQL query:

```
SELECT C.sid
FROM   Catalog C
WHERE NOT EXISTS (SELECT P.pid
                  FROM   Parts P
                  WHERE NOT EXISTS (SELECT CG.sid
                                    FROM   Catalog CG
                                    WHERE CG.sid = C.sid AND
                                           CG.pid = P.pid));
```

Write the relational algebra translation:

Question 15 [3 marks] Given the following SQL query:

```
SELECT S.sid
FROM Suppliers S
WHERE S.address = '1280 Main Street' OR
      S.sid IN (SELECT C.sid
                FROM Parts P, Catalog C
                WHERE P.pid = C.pid AND p.colour = 'red')
```

Write the relational algebra translation:

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